JIAN JIANG

Toronto, ON | 226-698-7306 | jiangjian9803@gmail.com | LinkedIn | Personal Website | GitHub

PROFESSIONAL SUMMARY

Adaptable engineering professional with 3+ years of experience in Python automation development and troubleshooting, plus hands-on Java/Spring Boot project experience. Fast learner and systematic problem-solver with strengths in software integration, debugging, and reliable workflow development.

TECH STACK

Languages	**Java**, **Python**, **SQL**, TypeScript, C/C++
Java Backend	**Spring Boot**, **Spring 6**, **REST APIs**, **JPA/Hibernate**, **Maven**
Web, Data & Messaging	React, **PostgreSQL**, **MySQL**, **RabbitMQ**, JSON, HTTP
Python Automation	Pandas, PyVISA, SCPI, CSV and log processing
Testing	**JUnit 5**, **Mockito**, pytest, automated testing, defect investigation
Tools & Integration	**Git**, SVN, Docker, UART, USB-C, GPIB, LAN

SOFTWARE DEVELOPMENT EXPERIENCE

UNIVERSITY OF GUELPH

Software Developer **Feb 2025 – Jul 2025**

Project: Wildfire Management System Modernization | *Faculty Advisor: Prof. Simon X. Yang*

Modernized a legacy .NET desktop application into a React and Spring-based web platform for wildfire-management workflows.

- **Architecture & Modernization:** Helped convert legacy desktop workflows into a web architecture connecting a React/TypeScript interface, Java/Spring Boot services, PostgreSQL persistence, and RabbitMQ messaging.
- **Backend & REST APIs:** Implemented backend functionality across controller, service, and repository components, including request validation, business processing, structured responses, HTTP status handling, and application exceptions.
- **Database Development:** Used JPA/Hibernate and PostgreSQL for entity mapping, queries, updates, and transactional data operations supporting wildfire-management records.
- **Messaging & Integration:** Supported RabbitMQ-based asynchronous workflows and aligned frontend request/response models with backend API contracts during integration.
- **Testing & Troubleshooting:** Developed JUnit 5 and Mockito tests, reproduced defects across application layers, verified fixes, and collaborated through Maven, Git, and code reviews.

PEGATRON KUNSHAN ELECTRONIC CO., LTD.

RF Automation Software Developer **May 2019 – Dec 2022**

Projects: Apple Vision Pro / iPhone 15 — New Product Introduction (NPI)

Developed and maintained Python automation code integrating production devices, RF instruments, and result-processing workflows across MLB and FATP stages.

- **Python Application Development:** Built reusable modules for configuration loading, DUT and instrument communication, workflow orchestration, result processing, logging, and resource cleanup.
- **DUT Integration Code:** Implemented UART and USB-C functions to transmit commands, parse responses, handle communication errors, and verify that devices entered the required operating state.
- **Instrument Integration Code:** Wrapped existing SCPI commands in Python functions and used PyVISA to configure, trigger, poll, and retrieve results from LitePoint and R&S instruments over GPIB and LAN.
- **Workflow & Validation Logic:** Coded reusable TX power, EVM, RX sensitivity, PER, BER, and RSSI workflows that coordinated device states, instrument operations, result parsing, and approved-limit checks.

- **Data-Processing Code:** Used Pandas to parse logs and CSV files, validate required fields, convert measurement values, handle invalid records, and extract failed results.
- **Debugging & Reliability:** Traced commands and responses to diagnose invalid states, delayed results, and timeouts; added state checks, completion polling, bounded retries, exception handling, and cleanup.
- **Code Quality & Ownership:** Used pytest for parsing and limit-checking logic and maintained code in SVN; owned the Python integration layer, not RF specifications, hardware, firmware, or release decisions.

ADDITIONAL EXPERIENCE

HUAWEI TECHNOLOGIES CO., LTD.

Technical Recruiter **Aug 2025 – Present**

- **Technical Automation:** Develop Python and OpenAI Codex-assisted workflows to consolidate Excel/CSV candidate data, remove duplicate profiles, validate public-source information, and prioritize candidates for human review against role-specific criteria.
- **Technical Recruitment:** Source and screen software engineering candidates and conduct initial qualification interviews.
- **Coordination:** Coordinate interviews with hiring managers and maintain candidate pipelines and recruitment records.

EDUCATION

Master of Applied Science in Engineering — University of Guelph, Canada | 2023–2025 | GPA: 3.85/4.0
Bachelor of Science in Electrical Engineering — Hubei University of Technology, China | 2015–2019 | GPA: 3.7/4.0